



The COVID-19 Impacts on Contextual Privacy Behavior in Kuwait

Tahani Hussain

System and Software Development
Kuwait Institute for Scientific Research
State of Kuwait
thussain@kisir.edu.kw

Alaa Ismail

Energy and Building Research Center
Kuwait Institute for Scientific Research
State of Kuwait
aljawad@mac.com

ABSTRACT

The medical, social and business disturbance associated with the COVID-19 coronavirus caused dramatical changes in the usage of mobile devices. People were forced during the lockdown to use different mobile-based context-aware application to cope their lives while staying home. Consequently, a typical user's privacy preferences and behavior were altered due to that fact that these applications at that moment are found to be an optimum solution to endure the predicament. This paper is aimed to assess how COVID-19 pandemic changes the mobile usage and contextual privacy behavior in Kuwait. Toward this goal, an information visualization system is developed to accumulate the user behavior dataset from their mobiles. The result shows the average mobile usage hours have increased by 60% and in total the user behavior changed by 27.9% in loosen their contextual privacy.

CCS CONCEPTS

•Security and privacy → Human and societal aspects of security and privacy → Privacy protections •Information systems → Information systems applications → Mobile information processing system •Human-centered computing → Visualization design and evaluation methods

KEYWORDS

Context-aware, COVID-19, Coronavirus, Kuwait, Privacy, Mobile, Data Visualization.

1 Introduction

The first twelve weeks of 2020 have forced many worldwide governments to implement a series of extensive lockdown measures due to COVID-19 pandemic to reduce the spread of the coronavirus among their populations. This lockdown enforces people to work from home, events to be postponed, schools to

shut their doors, airlines to cancel their flights, and shops and restaurants to close. Accordingly, the medical, social and business lifestyles were disturbed, and only essential services were allowed to continue operation, albeit with limitations. Such disruptions associated with the COVID-19 crisis caused people to be quarantined inside their houses, and many turned to use their mobile devices to stay connected. Moreover, during lockdowns many were forced to download and used different context-aware mobile applications to carry on with their lives while staying home. Consequently, they were forced to change their contextual privacy preferences and patterns of behavior during that time to permit them to complete their work, order groceries, pay bills online and many more with no exception to endure the difficulties caused by the crisis.

In 2019, a real-world dataset was collected from different countries including Kuwait to evaluate the efficiency of privacy behavior recognition and recommendation system as extension to previous work [1]. Motivated by that, this current work aims to illustrate an assessment on how COVID-19 pandemic has changed the contextual privacy behavior of mobile users in Kuwait is presented. Toward this, an information visualization system was developed to collect the contextual privacy behavior from the user mobiles during the pandemic time. A total of 33 volunteers from Kuwait who participated in the earlier 2019-dataset studies have agreed to install and use the developed information visualization system. The dataset before and after the pandemic are compared to assess the impact of COVID-19 on contextual privacy behavior in Kuwait.

The rest of the paper is structured as follows: Section 2 presents Kuwaiti government lockdown measurements; Section 3 shows the information visualization system; Section 4 discusses the comparison results and Section 5 concludes the paper.

2 Lockdown Measurements in Kuwait

On 24 February 2020, Kuwait announced the first confirmed case of COVID-19 [2]. While the first day recorded 5 confirmed cases, though on 5 August 2020, there were a total of 69,425 cases. On Saturday, 4 April, the Health Ministry reported the first death from the coronavirus pandemic. As of 5 August, a total of 468 deaths were reported with 87.73% recoveries (that is 60,906).

The first measure implemented by the Kuwaiti government was to suspend school classes for two weeks starting from 1 March 2020. Next, the government declared a public holiday from 12 to 26 of March. Afterwards, starting from 15 March, all commercial flights were suspended with an exception for cargo flights. Following that, a decision to proceed with a public partial curfew

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was activated on 22 March with non-curfew hours from 04:00 to 17:00 local time (LT). Moreover, a decision was made to order the closures of entertainment venues, stores, theaters, malls, gyms, restaurants, beauty parlors ... etc.

The hours of the daily partial curfew were amended to be from 16:00 to 6:00 LT starting from 6 April. The curfew was subsequently amended to start from 16:00 to 8:00 LT starting from 24 April. Furthermore, it was decreed that schools were to be closed until 4 August, and all other related education activities were suspended until further notice. Also, the imposed public holiday was indefinitely extended until further notice. All such lockdown measures were undertaken to help curb the spread of coronavirus among the population of Kuwait and force them to stay at home for their own safety. However, due to the alarming increase in the number of recorded new cases, the government decided to apply a full public curfew beginning from 10 and lasting until 30 of May. To ease the various hardships endured by the public, various online websites links and mobile application were established to enable booking appointments for necessary needs and essential services such including shopping at the supermarkets or scheduling appointments for hospital visits for emergency cases.

The government started taking steps towards a gradual return to normal life with a plan that consists of 5 phases [3]. The first phase was activated on 31 May by ending the full curfew and return to a partial curfew from 18:00 to 06:00 LT. Phase two that includes resuming 30% of government and public labor was activated on 30 June with amended curfew hours from 20:00 to 05:00. On 28 Jul, phase three was activated with increase in the labor capacity to reach 50%, amended curfew hours from 21:00 to 03:00 LT and resuming 30% of the commercial flights starting from 1 August.

3 Information Visualization System

Nowadays, information is considered to be a vital part of mobile users' life, with most of the information being generated, stored, accessed, and used for individual benefits. Thus, visualizing the information is considered to be very important for mobile users to provide clear insight to their huge and complex information [4]. Information visualization system, mainly, is aimed to provide interactive graphical representations of the users' information to help each individual to easily understand their information [5].

In this work, we developed an information visualization system to collect, analyze and illustrate the mobile usage information. This usage information includes the user privacy preferences and related behavior and activity while using context-aware applications. The system aims to provide a visual illustration of the information and store it as a dataset of variables. The developed system consists of two components: storage device and user interface. Figure 1 shows the framework of the developed information visualization system.

Initially mobile user inquires visualization of the related information from the user interface, which in turn forward it to the storage device and request for the result. The storage device processes and stores the information as a dataset of variables. After that, the dataset along with the analyses are sent to the user interface. Finally, the user interface will view the result using visual screens and pages for the user.

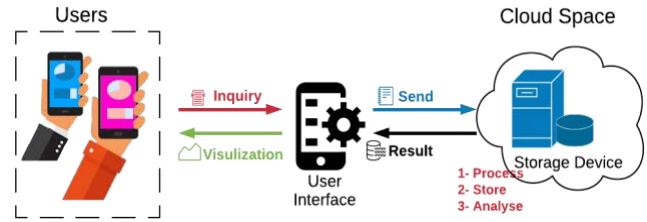


Figure 1. Framework of information visualization system.

3.1 User Interface

The user interface component is an iPhone Operating System (iOS) mobile application that is developed using Swift programming language version 5.2 [6]. It acts as an intermediary environment between the mobile user and the storage device component. It is designed to manage the information using interactive graphical pages. The application consists of three main screens to provide visualization of the commonly mobile usage information considered in this work, (see Figure 2).



Figure 2. User interface screenshot.

Figure 2(a) shows *Today Top* screen that is aimed to illustrate the mobile's top usage metrics interim of active hour, application, category and power usage. The *Current Usage Metric* screen shown in Figure 2(b) is meant to provide interactive and graphical metric for the mobile current power usage and categories usage pie chart respectively. The *Privacy Activity* screen, Figure 2(c), is developed to show the contextual privacy user's activity during the interim week of allowing, denying and approximating the context along with change in previous recorded activity.

4 Results

The assessment presented in this work relies on data collected from 33 volunteers from Kuwait (17 Males and 16 Female) of age groups ranging from 20 years old to subjects in their 60s, see Figure 3. As illustrated, the age/gender variety is fairly random Chi-Square (Chi-SQ) test = 0.176 with 4 Degrees of Freedom (DOF). A 24% of the volunteers are diploma holders, 67% are college graduates while 9% claim a graduate degree. The test of randomness for the gender/education level distribution yields a Chi-Square test = 0.611 (DOF=2).

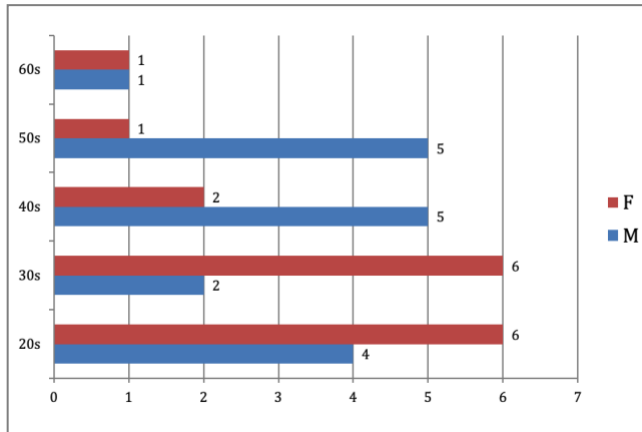


Figure 3. Gender and age distributions of the participants.

Generally, an investigation of the individuals mobile usage habits post the onset of COVID-19 in comparison to their habits as recorded in 2019 illustrates that the entire group has unanimously amplified their mobile usage. In 2019, the average amount of usage by all the participants was 5.5-hours, with the minimum duration of 2.7-hours and a maximum of 9.2-hours. Post COVID-19 the average has increased to 8.8-hours, with a minimum of 3.95-hours and a maximum of 12.2-hours. The participant with the smallest increase (23.4%) was a 25-yr old male, while the largest increase (113.0%) belonged to a 27-yr old male. We found that the average amount of increase for all the participants was 63.2%. About 31% of the subjects have increased their mobile usage by 50%. And finally, the increase in mobile usage is relegated to the users age group (from 20s to 60s).

4.1 Mobile Usage Behavior

Table 1 summarizes the increase in time spent in mobile usage per each application category. It is noticed that hourly use in all categories have increased in 2020 after the onset of COVID-19, and the greatest increases in usage in specific categories are those observed for “Education” (246.3% increase), followed by “Entertainment” (123.9%), “Media” (80.0%), “Shopping” (97.8%) and “Fitness” (78.3%). Some of these categories, namely “Gaming”, Entertainment” along with “Social Media” (which had a minimal increase of 17.3%) were already the dominant categories in use before the inception of COVID-19. The “Education” category experienced the greatest increase even though it happened to be a minor class pre-COVID-19. Increases in such categories epitomizes the change in social behavior imposed upon the population by the imposition of curfew and quarantine measures intended to curb the spread of COVID-19 among the population.

Category	2019	2020	% Increase
Education	4.97	17.21	246.3%
Entertainment	25.91	58.01	123.9%
Shopping	20.16	39.87	97.8%
Media	9.07	16.33	80.0%
Fitness	12.51	22.31	78.3%
Gaming	45.38	58.41	28.7%
Medical	9.07	11.66	28.6%
Social Media	36.29	42.57	17.3%
Utilities	16.49	17.16	4.1%
Other	1.62	6.54	303.7%

Table 1. Average increases in mobile usage per category.

Figure 4 shows the total time spent by all the volunteers in using their mobile per week for each daily hour during the curfew weeks (22 Weeks). The curfew hours for every week are highlighted by gray color.

Time	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
0:00	6.2	5.6	6.1	7.1	7.5	7.8	6.6	9.8	11.0	15.0	13.1	12.2	3.8	3.0	13.7	11.8	9.3	10.0	8.9	8.9	7.2	7.9
1:00	4.7	5.2	6.1	6.7	7.4	8.3	8.8	9.9	16.8	13.1	13.2	12.4	8.0	6.2	2.7	8.7	7.6	8.3	5.9	6.1	6.1	9.9
2:00	3.2	2.9	3.7	3.9	5.7	6.6	6.2	6.8	13.1	13.2	12.4	8.0	6.2	2.7	8.7	7.6	8.3	5.9	6.1	6.1	9.9	6.5
3:00	0.8	2.1	5.8	5.0	6.2	6.1	6.1	7.0	14.7	15.0	10.0	6.8	5.8	3.9	10.4	8.5	10.9	3.3	5.2	3.9	6.2	3.9
4:00	2.3	1.0	3.9	3.4	2.9	3.1	5.8	6.9	10.0	13.3	12.3	8.9	6.6	4.0	9.2	11.0	7.5	2.8	4.0	3.1	9.2	3.8
5:00	3.0	1.0	5.8	3.0	3.8	3.1	6.5	7.3	7.3	11.0	13.9	9.8	7.9	4.2	3.9	7.7	3.4	5.4	8.1	3.0	5.2	3.8
6:00	4.7	2.9	0.8	0.3	7.1	3.3	6.1	8.3	6.5	9.3	8.6	6.6	6.1	7.4	5.9	4.4	6.1	6.1	5.7	5.2	3.8	2.8
7:00	5.1	3.6	3.0	3.0	3.3	3.4	2.9	3.8	3.3	9.2	5.8	7.8	3.2	9.1	4.3	5.6	3.8	8.3	4.0	6.5	3.3	3.2
8:00	6.7	6.7	2.3	2.6	2.8	3.1	3.7	3.0	3.9	3.4	8.2	8.8	3.2	3.9	4.0	7.2	4.2	4.4	7.8	6.5	4.2	3.1
9:00	6.6	6.4	0.4	4.1	5.7	4.7	4.2	2.0	3.1	2.4	3.7	6.9	4.8	5.4	7.3	8.4	5.3	9.2	5.3	8.3	6.2	3.9
10:00	6.7	6.4	3.6	2.9	5.7	3.9	4.1	1.7	2.4	1.0	3.8	3.6	3.8	6.1	5.2	4.1	5.7	7.2	6.1	5.2	8.9	6.2
11:00	6.8	6.4	7.9	3.1	1.5	4.2	5.1	1.4	1.7	0.9	3.0	5.9	3.7	6.7	6.3	8.0	5.8	3.9	6.9	7.5	6.6	4.3
12:00	6.6	5.2	4.2	3.1	1.9	4.2	6.1	2.3	1.8	0.7	0.6	2.5	3.0	8.3	7.6	7.5	3.7	4.6	7.2	9.2	5.9	2.0
13:00	6.2	3.7	5.5	2.4	3.0	6.6	3.2	2.3	2.2	2.1	3.1	0.8	3.2	8.2	8.9	3.5	0.5	7.1	7.2	3.8	3.8	3.5
14:00	4.5	3.0	3.0	3.0	3.1	6.6	3.4	3.0	0.3	4.7	5.9	0.4	4.4	7.0	6.7	6.6	4.9	6.3	4.5	6.1	3.7	6.5
15:00	7.1	6.0	6.5	3.5	3.4	3.0	5.9	2.7	0.6	5.7	6.1	0.6	3.9	7.7	5.5	5.2	3.3	5.7	4.1	3.7	6.1	4.0
16:00	7.0	7.1	8.3	5.7	3.9	5.2	6.5	3.1	0.5	5.9	4.2	3.0	5.2	7.6	3.9	7.6	3.6	8.8	4.0	3.1	3.9	5.2
17:00	11.5	9.5	9.6	9.1	8.5	6.1	7.3	3.3	1.0	7.0	3.9	3.9	4.6	8.7	7.2	6.6	4.1	3.7	4.1	1.6	6.3	6.6
18:00	6.6	12.7	12.8	10.0	10.0	6.9	9.7	3.0	5.5	8.8	5.8	4.1	6.8	11.3	6.6	7.2	5.9	3.7	7.9	5.2	8.3	8.3
19:00	6.9	14.4	11.6	14.5	14.8	7.8	11.2	6.1	8.3	10.1	6.1	5.8	9.9	11.0	7.9	7.1	3.9	5.4	10.8	6.5	6.1	8.6
20:00	11.5	15.8	6.5	9.9	13.9	6.6	7.2	7.1	11.9	11.8	6.6	5.8	11.0	7.7	11.0	8.3	9.2	10.4	9.1	6.8	4.9	9.1
21:00	4.0	12.6	11.5	12.6	12.4	9.2	9.2	9.9	12.4	11.7	7.9	6.2	8.3	11.1	11.1	10.8	10.0	9.6	10.3	9.3	8.3	10.8
22:00	3.8	12.6	12.0	6.7	9.2	6.2	8.4	10.5	12.7	14.0	7.0	6.6	6.1	11.8	12.4	10.0	13.1	7.6	10.8	6.3	9.4	11.4
23:00	7.3	9.8	9.6	12.0	9.2	9.6	8.6	9.9	15.5	14.1	10.0	8.3	6.5	12.9	14.2	9.2	10.2	8.7	11.7	9.2	9.3	12.0

Figure 4. Mobile usage hours over 22 curfew weeks.

Figure 5 sums up the hours of activities spent both in the open free time during the curfew period, as well the activities registered during the curfew hours over the entire curfew period.

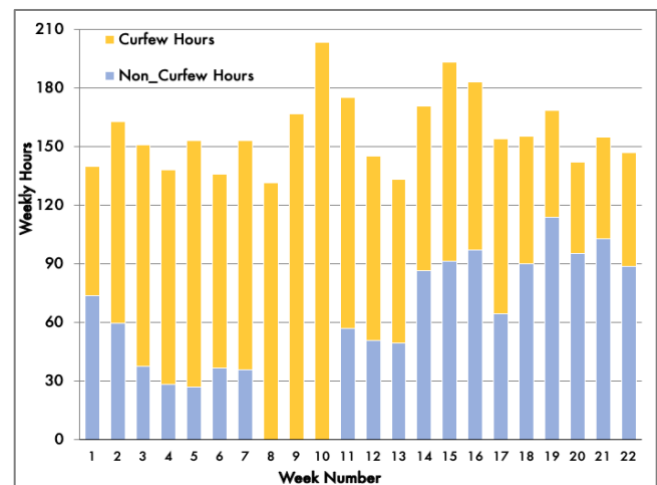


Figure 5. Weekly hours of activity under curfew.

In both figures it should be noted that the public had spent weeks 8, 9 and 10 under a total lockdown, hence the absence of a non-curfew summation in Figure 5 for those weeks.

Activities within the curfew hours increased progressively over the weeks leading to that 3-week period. After week 10 the ratio of the aggregate of activities within the curfew hours to the total activity within each week started to decline steadily, reaching only 39.5% on week 22. Most of the users tended to spend their curfew hours using their mobile with 63.42% of the mobile usage hour were during the curfew time. Moreover, it can be noticed that users during the daylight hours spend less time using their mobile. Thus, users were forced to change their sleeping habit and stay overnight using their mobiles during the total lockdown.

It should be noted that week 10 witnessed the leading number in terms of hours of all the weeks within the period. This is due to the fact that time period coincided with the festivities associated with “Eid al-Fitr”, an event marking the end of the Islamic Holy month of Ramadan. This occasion is usually associated with massive public celebrations and heightened social activities, this year however it was observed under a complete lockdown.

The increases noted in weeks 15 and 16 are mostly due to an increase in activities associated with relaxing the curfew measures.

4.2 Contextual Privacy Behavior

The recorded activities for all users have been compiled and binned to 4 types: *Allow*, *Deny*, *Approximate* and *Change*, and the 2020 results were compared to the pre-COVID-19 behavior in 2019. Figure 6 illustrates the overall percentage differences for all activities between the two years.

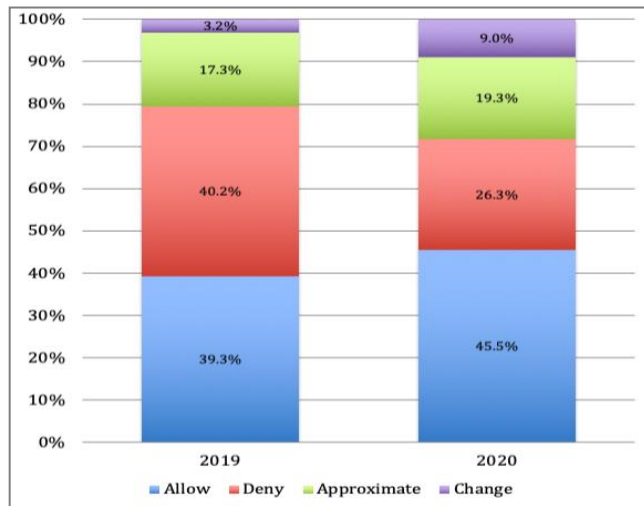


Figure 6. Overall percentage differences of privacy behavior.

It is noticed that all of the activity types have increased in the 2020 summation, except for the *Deny* activity, which has dropped from 40.2% to 26.3%. Expressly, users have changed their privacy behavior by releasing 13.9% of the context that they have denied pre-COVID-19. Altogether, the combined change in all activities amounts to a decline of 27.9% in the loosening the contextual privacy behavior has been captured. That is users were willing to release more of their context in trade off using the

context-aware application to cope with their lives during the COVID-19 protection measures. Furthermore, the accumulated percentage differences have been investigated based on the nature of the activity corresponded to the application categories (e.g., Medical, Education, Shopping, Social Media and Utilities), and the results are summarized in Table 2 (*Allow* and *Deny*) and Table 3 (*Approximate* and *Change*).

Category	Allow		Deny	
	2019	2020	2019	2020
Medical	39%	49%	8%	4%
Education	36%	43%	10%	5%
Shopping	15%	21%	45%	37%
Social Media	16%	22%	36%	32%
Utilities	22%	160%	41%	115%

Table 2. The privacy behavior (Allow and Deny) per category.

What was noticed in Table 2 is that universally there was a decline in the *Deny* activity in 2020 in comparison with 2019, which most prominently observed in the “Medical” and “Education” categories. Moreover, a massive change in the *Allow* behavior can be noticed in the “Utilities” category which can be justified by the fact that the applications falls in this category were necessitous by the users during the lockdown. Similarly, the impact of the lockdown can be noticed on the *Approximate* and *Change* user behavior for the same category, see Table 3. Thus, user’s privacy behavior was altered extremely to endure the predicament during the lockdown.

Category	Approximate		Change	
	2019	2020	2019	2020
Medical	19%	11%	35%	36%
Education	20%	12%	34%	40%
Shopping	31%	26%	9%	16%
Social Media	39%	36%	9%	10%
Utilities	26%	114%	11%	111%

Table 3. The privacy behavior (Approximate and Change) per category.

The weeks of observations are presented in a chronological order of monitoring, and largest number of cases was observed in the 7th week of monitoring (just before the 3-week complete lockdown period). The lowest declines in the percentages, which were all in the negative ranges, were most prominently observed in the categories: “Education”, “Shopping” and “Social Media”, and such declines were observed within the first 6 weeks of monitoring the various privacy behaviors in 2020. On the other hand, the largest increases ranging from 21.5% to 31% were observed within the last 3 weeks of monitoring, and they were specific to the “Entertainment”, “Fitness”, “Shopping” and “Social Media”. Individual plots of application categories vs. weekly percentage changes over the 13-week monitoring period are presented in Figure 7.

Investigating the patterns of change within each category yields a few interesting observations. For example, there was a sudden jump in the percentage and increase afterwards in the “Social Media” category centered on week 7. A similar jump was also observed in the “Utility” category, except with a steady decline after the 7th week. The category: “Other” experienced a steady

increase until peaking in the 9th week, and the percentages steadily declined afterwards. The categories: Entertainment, Shopping, Fitness and to some extent Medical have all experienced a steady increase over the entire period of monitoring. There were no discernable patterns of behavior associated with the “Media” and “Education” categories.

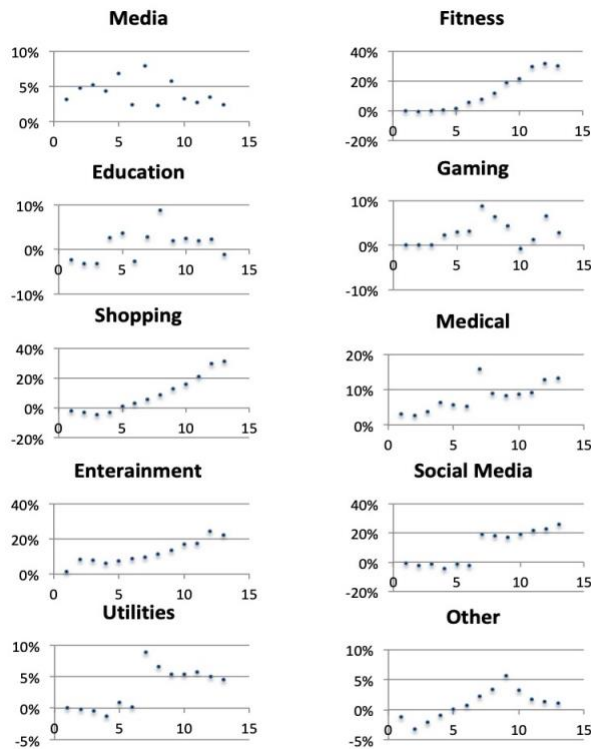


Figure 7. Change in the percentages of patterns of user privacy behavior for the various categories over 13 weeks.

From the results shown in Sections 4.1 and 4.2, we found that the disturbance and lockdown measurements associated with the COVID-19 coronavirus led to major increase in the usage of the “Education”, “Shopping”, “Entertainment”, “Fitness” and “Social Media” applications. However, with the increasing of the lockdown measurements the noticeable increase was in “Entertainment”, “Fitness” and “Gaming” applications. In the first week of partial public curfew in Kuwait, the weekly mobile usage rate increased by 74% compared to the weekly usage rate pre-COVID-19 and 35% of it was in using “Gaming” and “Social Media” application. In the first week of the full public curfew, 10-17 May 2020, there was 88.1% increase compared to the weekly average during the partial curfew. Afterword, due to the fact that the users are spending more time at home, we noticed huge increase in using “Fitness” and “Medical” application. Users were interested in applications to help them in relieving stress, relaxation, workout and exercising at home with increase rate of 43%. Two months after announcing the first confirmed case of COVID-19 in Kuwait, mobile users have loosened their privacy restrictions by 16.5%.

As a conclusion we found that mobile user behavior interim of usage hours and contextual privacy preferences have altered during the COVID-19 pandemic. We foresee a shift in the user’s

habits and perceptions, which will continue to change towards spending more time using their mobile devices and exploring more applications. Therefore, we will witness a vast evolution in the mobile technologies and more growth in number of applications.

7 Conclusions

In this paper, assessment on the impact of the COVID-19 pandemic on the contextual privacy behavior of mobile users in Kuwait is presented. Two datasets of information collected from 33 mobiles in 2019 as well as after the onset of the global pandemic in 2020 are used for the assessment methodology. Moreover, an information visualization system to collect the contextual privacy behavior from the user’s mobiles is proposed in this work.

The result shows that during the COVID-19 pandemic the average mobile usage hours have increased by 60%. Moreover, application categorized as “Education” experienced the greatest increase even though it happened to be a minor class pre-COVID-19. When it comes to change in the behavior of various types of activities over a 13-week period during the COVID-19 crisis, the categories related leisure online activities (e.g., Entertainment, Fitness and Shopping) have demonstrated a steady increase over that period. Overall, we found that users were willing to release 27.9% more of their context in trade off using the context-aware application to cope their lives during COVID-19.

It should be interesting to collect future observation at a post-COVID19 period and compare the results with the outcome of this present work.

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